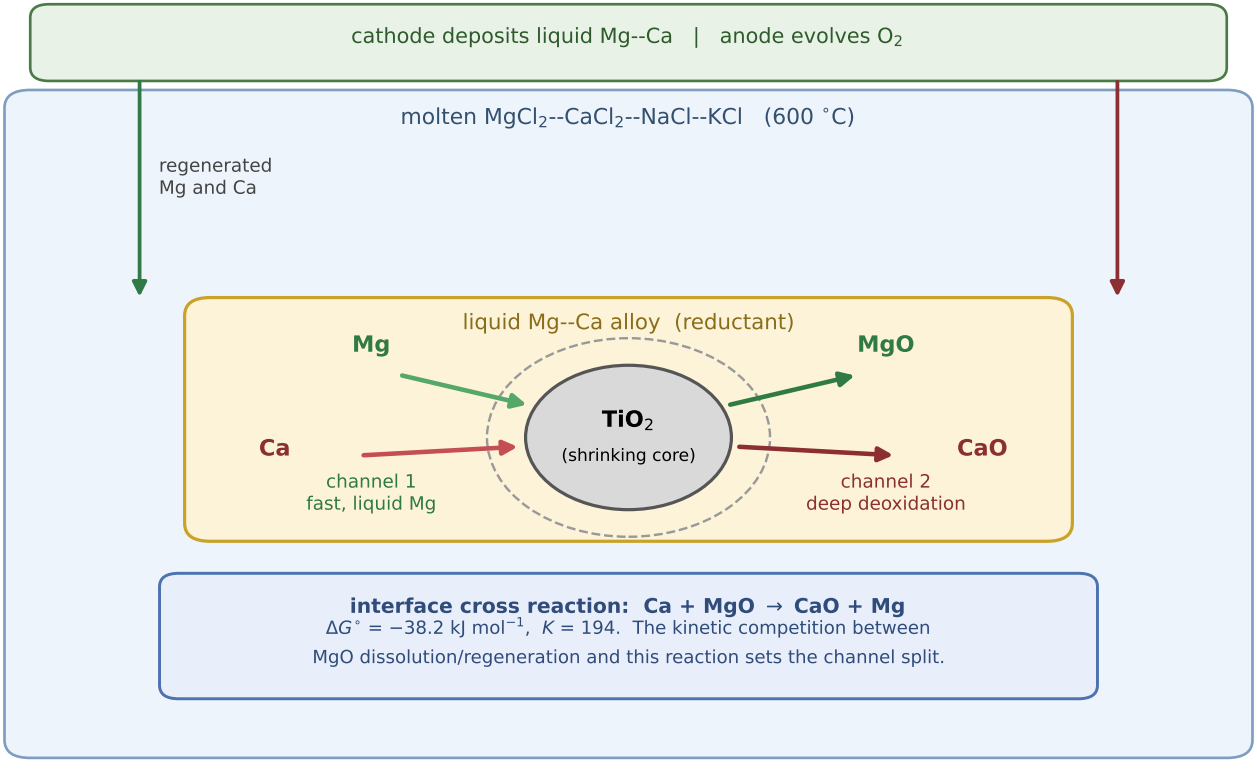

Net reaction: $TiO_2 \rightarrow Ti + O_2$ (Ca is the net oxygen carrier)

Mg makes the reductant liquid at 600 °C, where both pure metals are solid